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A. SOFTWARE DOMAIN
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**PS 1: DisruptionIQ - From Delivery Delays to Recovery Plans**
**Domain: Logistics and Supply Chain Management**

**Problem:**
Logistics teams get fragmented updates about delayed shipments, supplier disruptions, inventory shortages and delivery commitments. Spotting a late shipment is easy. Knowing which orders are affected, how serious it is and what to do about it is hard.

**Challenge:**
Build a platform that combines shipment, inventory, supplier and order data to find downstream risks, prioritise affected customer orders and recommend recovery plans.

**Must include:**
- Data import from CSV/JSON and at least one simulated carrier webhook
- Linked data model: suppliers, shipments, warehouses, inventory and orders
- Detection of late shipments, supplier delays and stock shortages, with impact analysis
- Risk ranking of orders with an explanation of the factors
- AI-assisted recovery options (reallocate stock, expedite, alternative supplier, split order) based on available data
- Approve / reject / modify workflow, with execution simulated only after confirmation
- Notifications and a full audit trail

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- Dashboard of active disruptions, affected orders, pending approvals and recovery progress
- Backend-enforced role-based access (Operations Analyst, Supply Chain Manager, Warehouse Operator, Procurement Approver, Administrator), with no self-approval where separation of duties applies

**Constraints:**
- Persistent database required
- Generate your own synthetic data (no dataset provided)
- Demonstrate at least two disruption scenarios
- AI for summaries and recommendations; inventory rules, deadlines and permissions must use deterministic logic
- Handle duplicate events, invalid data and conflicting updates safely